



SIMATS
ENGINEERING
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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)



AI-Driven Scholarship Eligibility and Application Recommendation Portal

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Problem Background

- Many scholarship portals provide only basic search and application services.
- Students find it difficult to identify scholarships that match their profile.
- Existing systems often provide limited or non-personalized recommendations.
- Students still spend time checking eligibility for multiple scholarships.
- An AI-driven portal can provide accurate eligibility prediction and personalized scholarship recommendations.



Problem Statement

- Existing scholarship portals do not always provide personalized recommendations.
- Eligibility checking across different scholarships can be time-consuming.
- Students may miss suitable scholarships due to complex eligibility criteria.
- Managing applications through different portals is inconvenient.
- An AI-driven recommendation portal can simplify scholarship discovery and improve application success.



Objectives

- To create a student profile that stores academic and personal details.
- To predict scholarship eligibility based on the student's profile.
- To recommend suitable scholarships using AI-based analysis.
- To simplify the scholarship application process through a single platform.
- To help more students apply for scholarships and reduce missed opportunities.

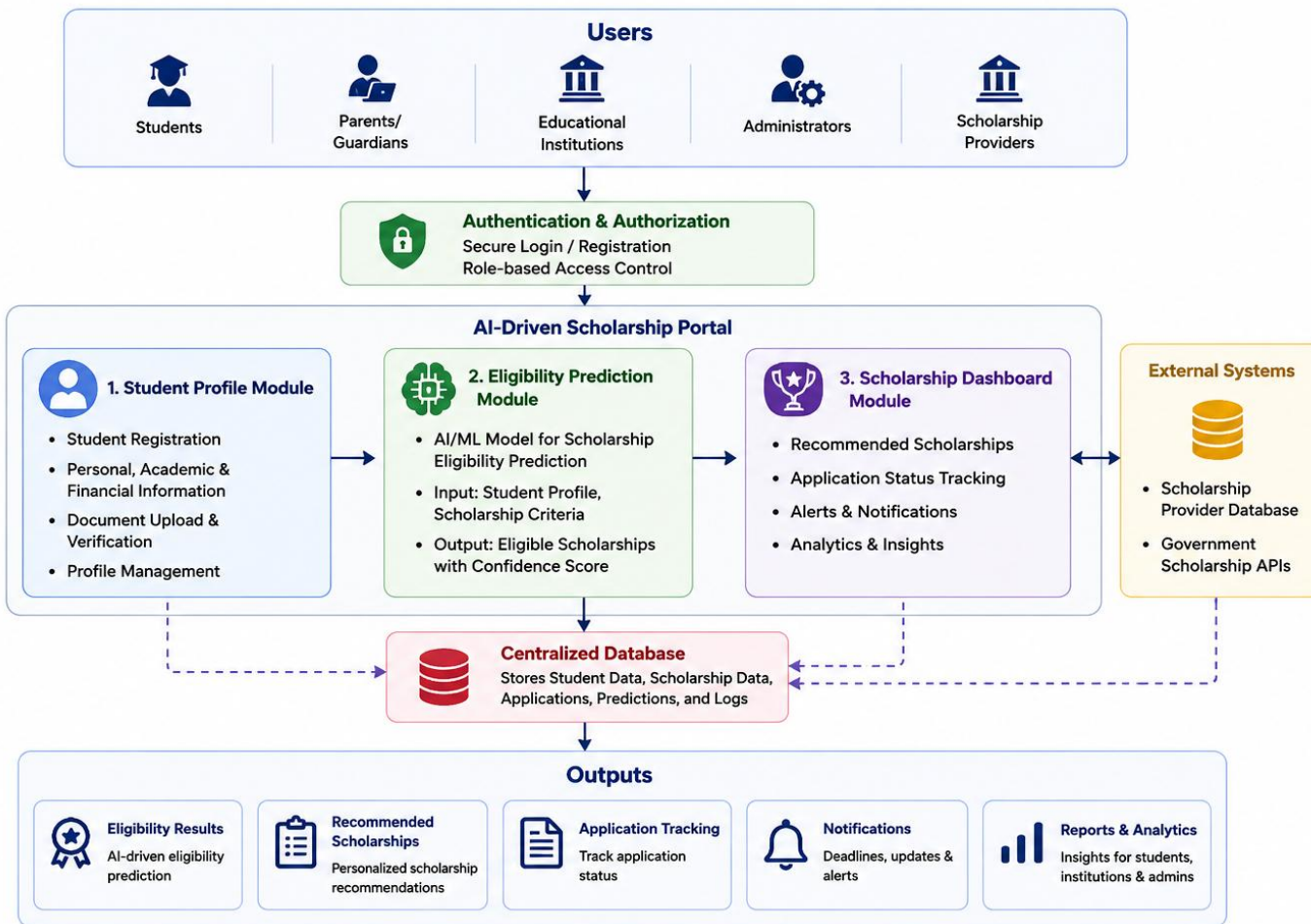


Literature Review

S.No	Title of the paper	Publisher and Published Year	Components /Techniques used
1	AI-Based Scholarship Recommendation System	IEEE, 2025	Machine Learning, Student Profile Analysis, Recommendation System
2	Intelligent Student Scholarship Management System	Springer, 2025	Eligibility Prediction, Web Application, Database Management
3	Machine Learning for Educational Decision Support	ScienceDirect, 2024	Classification Algorithms, Student Data Analysis, Prediction Models
4	Smart Scholarship Portal Using Artificial Intelligence	IEEE, 2024	AI Recommendation Engine, User Authentication, Dashboard
5	Online Scholarship Management System	Elsevier, 2025	Application Tracking, Notification System, Database Integration

Architecture

AI-Driven Scholarship Eligibility and Application Recommendation Portal – Architecture





Module 1: Student Profile Module

- **Collects Personal Details** – Stores the student's name, age, gender, and state.
- **Captures Education Details** – Records education level, course, college, year, and academic marks.
- **Stores Financial Details** – Collects family income and other basic financial information.
- **Identifies Special Eligibility** – Records category, PwD status, minority status, first-generation status, and other conditions.
- **Creates a Complete Student Profile** – Combines all details to prepare the student for accurate scholarship eligibility checking.



Module 1: Student Profile Module

STEP 1 • DEMOGRAPHICS & IDENTITY

Personal Details

Provide identification details to match state quotas and category reservation endowments.

FULL LEGAL NAME *

Bala

EMAIL ADDRESS *

bala12345@gmail.com

PHONE NUMBER

+91 9876543210

DATE OF BIRTH

15-06-2004

GENDER

Male

STATE / UT OF DOMICILE *

Maharashtra

SOCIAL RESERVATION CATEGORY *

General / Unreserved

PERSON WITH BENCHMARK DISABILITY (PWD)

☐ Yes (40%+ certified) ☒ No

Cancel

Save & Continue to Education →

STEP 2 • ACADEMIC STREAM

Select Your Current Education Level

Your selection dynamically determines the relevant academic marks and eligibility rubrics.



High School / Class 10

Class 9-10 secondary schooling & NTSE/NMMS



Higher Secondary / Class 12

Class 11-12 junior college, PCM/PCB/Commerce/Arts



Undergraduate Degree

B.Tech, B.Sc, MBBS, B.Com, B.A. 3-4 year programs



Postgraduate / Master's

M.Tech, M.Sc, MBA, M.A, MD advanced study



Polytechnic / Diploma

Technical & professional 2-3 year diplomas



Vocational / ITI

Craftsmen Training Schemes & NTC trade programs

← Back

Continue to Academic Details →

STEP 3 • UNIVERSITY & COURSE DETAILS

Undergraduate Records

Enter your college's degree's current year's and cumulative CGPA.

COLLEGE / UNIVERSITY NAME *

SIMATS Engineering

DEGREE / COURSE *

B.Tech / B.E. (Engineering)

BRANCH / MAJOR *

CSE

CURRENT YEAR *

3rd Year

CUMULATIVE CGPA (OUT OF 10) *

8.5

EXPECTED GRADUATION *

2028

← Back

Save & Continue to Eligibility →

STEP 6 • PROFILE REVIEW

PROFILE COMPLETION

Profile Summary Review

100%



Personal Information

Edit

Applicant: Bala

State: Karnataka

Category: GENERAL

PwD Status: No



Academic Records

Edit

College & Course:

B.Tech in CSE

Cumulative CGPA:

8.5 / 10.0

Institution:

SIMATS Engineering



Financial & Special Criteria

Edit

Annual Income: ₹2,00,000

Special Quotas: 0 selected

Back

Check My Eligibility with AI



Module 2: Eligibility Prediction Module

- **Analyzes Student Data** – Uses the student's academic, financial, and eligibility details.
- **Uses Decision Tree** – The AI model predicts the student's scholarship eligibility.
- **Checks Scholarship Criteria** – Compares the profile with scholarship requirements.
- **Generates Match Score** – Gives a score based on how well the student matches each scholarship.
- **Explains the Result** – Shows why the student matches or what information is missing.



Module 2: Eligibility Prediction Module

TARGETED GRANT MATCH

Eligibility Analysis Complete

Eligibility Analysis Complete: 7 Verified Eligible Schemes, 0 Possible Matches. ML Signal:
HIGH_MERIT_NEED (Score: 91%)

Eligible Scholarships
7
₹5,61,000 total funding

Possible Matches
0
Requires document verification

Not Eligible
27
Criteria threshold not met

ELIGIBLE ALL • UNDERGRADUATE

91% Match

Reliance Foundation Undergraduate Scholarship

₹2,00,000

per academic year

Reliance Foundation

Reliance Foundation CSR
Verified official government / institutional scholarship

[View Source](#)

Why You Qualify

Education: UNDERGRADUATE | Academic: 80.8% | Income: ₹200000 |
Course: B.Tech | ML Decision Tree: HIGH_MERIT_NEED

Required Documents

Class 12 Board Marksheets with 60%+ aggregate, Proof of admission in 1st year regular UG degree, Annual Family Household Income Proof (Preference ≤ ₹2.5L; Cap ≤ ₹15L), Aadhaar Card and Bank Passbook, Online Aptitude Test Scorecard

About this Scheme

Prestigious merit-cum-means scholarship for academically excellent first-year undergraduate students pursuing any regular degree stream in India with leadership potential.

Application Process:

Apply on Reliance Foundation scholarship portal. Complete 60-minute online cognitive & verbal aptitude test. Selection on test + academic merit.

Application Deadline:

2025-10-15

[Save Scholarship](#)

[Apply Now on Official Portal](#)

Verified Scholarship Evaluations

Explore categorized grants based on explicit requirements.

Eligible (7)

Possible Matches (0)

Not Eligible (27)

NOT ELIGIBLE ALL • UNDERGRADUATE

46% Match

AICTE Pragati Scholarship for Girl Students in Technical Education

All India Council for Technical Education (AICTE), Ministry of Education

₹50,000

Satisfied Criteria

- Education level matches (UNDERGRADUATE)
- Academic score (80.8%) meets minimum (50.0%)
- Family income (₹200000) is within ceiling (₹600000)
- Open to all social categories
- Open to all States and UTs across India
- Course (B.Tech) aligns with eligible specializations

Unmet Requirements

- Gender: exclusively for [FEMALE] (yours: MALE)

Deadline: 2025-12-31

[View Full Requirements & Apply](#)

NOT ELIGIBLE GENERAL • UNDERGRADUATE

46% Match

Maharashtra Rajarshi Chhatrapati Shahu Maharaj Shikshan Shulkh Shishyavrutti (EBC)

Directorate of Higher Education (DHE) & DTE, Govt of Maharashtra

₹50,000

Satisfied Criteria

- Education level matches (UNDERGRADUATE)
- Academic score (80.8%) meets minimum (50.0%)
- Family income (₹200000) is within ceiling (₹600000)
- Open to all social categories
- Course (B.Tech) aligns with eligible specializations

Unmet Requirements

- State: restricted to [Maharashtra] (your state: Karnataka)

Deadline: 2025-11-30

[View Full Requirements & Apply](#)

NOT ELIGIBLE SC • UNDERGRADUATE

45% Match

Top Class Education Scheme for SC Students in Premier Institutes

Ministry of Social Justice and Empowerment, GoI

₹2,00,000

Satisfied Criteria

- Education level matches (UNDERGRADUATE)
- Academic score (80.8%) meets minimum (60.0%)
- Family income (₹200000) is within ceiling (₹600000)

Unmet Requirements

- Category: restricted to [SC] (yours: GENERAL)



Module 3: Scholarship Dashboard Module

- **Finds Suitable Scholarships** – Displays scholarships that match the student's profile and eligibility.
- **Provides Scholarship Details** – Shows eligibility criteria, benefits, required documents, deadline, and application information.
- **Allows Scholarship Application** – Helps students access the official application and record their application status.
- **Tracks Applications** – Keeps track of applications such as Applied, Under Review, Approved, or Rejected.
- **Updates Scholarship Information** – Keeps scholarship details, deadlines, and application status available in one place.



Module 3: Scholarship Dashboard Module

UNDERGRADUATE • PAN-INDIA • CATEGORY GENERAL

Welcome back, Bala!

All metrics and applications are synchronized live with your Supabase profile.

22%

Profile Completion
Incomplete fields present
[Edit Profile Details --](#)

ELIGIBLE FUNDING POOL
₹5,61,000
Across verified eligible grants

ELIGIBLE SCHOLARSHIPS
0
[View Eligible Grants --](#)

ACTIVE APPLICATIONS
0
[Track Application Status --](#)

SAVED SCHOLARSHIPS
0
[Explore Verified Catalog --](#)

Academic Performance Snapshot

Update Marks

DEGREE & COURSE
B.Tech in CSE

CUMULATIVE CGPA
8.50 / 10.0

COLLEGE / INSTITUTE
SIMATS Engineering

CURRENT YEAR & GRAD
3rd Year (2028)

Top Verified Matches

View All (4)

Strict requirement-evaluated scholarships from National & State Portals.

67% Match

Social Welfare & Women Empowerment Dept, Govt of Tamil Nadu

Tamil Nadu Moovalur Ramamirtham Ammaiyar Higher Education Assurance (Pudhumai Penn)

Potential Match (Data Needed): Gender not specified, State of domicile not specified | MODERATE_MERIT_NEED

₹12,000
per year

View & Apply

62% Match

Department of Higher Education, Ministry of Education, GoI

Central Sector Scheme of Scholarships (CSSS) — College & University

Potential Match (Data Needed): Academic marks/percentage not provided, Family income not entered in profile | MODERATE_MERIT_NEED

₹20,000
per year

View & Apply

62% Match

Kendriya Sainik Board (KSB), Ministry of Defence, GoI

Prime Minister's Scholarship Scheme (PMSS) for Wards of Ex-Servicemen

Potential Match (Data Needed): Academic marks/percentage not provided | MODERATE_MERIT_NEED

₹36,000
per year

View & Apply

62% Match

Department of Science and Technology (DST), Ministry of Science & Tech, GoI

INSPIRE Scholarship for Higher Education (SHE) — Natural Sciences

Potential Match (Data Needed): Academic marks/percentage not provided | MODERATE_MERIT_NEED

₹80,000
per year

View & Apply

19-08-2026

12



Existing vs Proposed System

Existing System	Proposed System
Provides a list of scholarships.	Recommends scholarships based on the student's profile.
Eligibility checking is limited or manual for many schemes.	AI predicts eligibility automatically.
Recommendations are not fully personalized.	Personalized recommendations using AI.
Application tracking is limited.	Complete application tracking with status updates.
Limited notification support.	Timely notifications and reminders.
Basic search features.	Smart filtering and AI-based recommendations.



Future Scope

- **Expand Scholarship Coverage** – Include more central, state, university, and private scholarships across India.
- **Improve AI Predictions** – Use larger and regularly updated datasets to improve eligibility prediction.
- **Automatic Scholarship Updates** – Detect changes in scholarship eligibility, deadlines, and availability.
- **Mobile Application** – Develop a mobile app so students can access the platform anywhere.
- **Smart Notifications** – Send personalized alerts for new scholarships and approaching deadlines.



Conclusion

- **Provides a complete scholarship platform** by combining student profiling, eligibility analysis, recommendations, and application tracking.
- **Uses a Decision Tree** to analyze student information and predict suitable scholarship opportunities.
- **Reduces student effort** by simplifying scholarship search and eligibility checking.
- **Provides personalized recommendations** based on academic, financial, and personal details.
- **Helps students manage applications** and keep track of scholarship deadlines and status.
- **Creates a more accessible and organized scholarship discovery process** for students across India.



References



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